

# Chi-square tests and one-way ANOVA

## Overview

For aggregate contingency tables — the kind that sit behind the Doob  $\chi^2$  family in the MRM framework — MORIE exposes `chi_square_test()` and the companion effect-size helpers (Cramer's V, omega-squared) as first-class functions.

## A 2x3 contingency table

```
library(morie)
tab <- matrix(c(20, 10, 30,
                15, 25, 35), nrow = 2, byrow = TRUE,
              dimnames = list(group = c("treated", "control"),
                              outcome = c("A", "B", "C")))

result <- chi_square_test(tab)
result$statistic
#> NULL
result$p_value
#> [1] 0.05145575
result$df
#> [1] 2
```

`chi_square_test()` returns the Pearson  $\chi^2$ , the asymptotic p-value, and the degrees of freedom in a tidy list.

## Effect size: Cramer's V

```
v <- cramers_v(tab)
v
#> [1] 0.209657
```

Cramer's V scales the chi-square statistic to a 0–1 association measure that is more interpretable than the raw statistic.

## One-way ANOVA + omega-squared

For continuous outcomes across more than two groups, the one-way ANOVA pattern in R is `aov(y ~ g, data = ...)`. MORIE exposes `omega_squared()` as the less-biased counterpart to eta-squared:

```

set.seed(2)
n      <- 90
group  <- rep(c("A", "B", "C"), each = n / 3)
y      <- rnorm(n) + ifelse(group == "C", 0.6, 0)

fit <- stats::aov(y ~ group)
fsum <- summary(fit)[[1]]
fstat <- fsum$`F value`[1]
df_b <- fsum$Df[1]
df_w <- fsum$Df[2]

omega_squared(f_stat = fstat, df_between = df_b, df_within = df_w, n = n)
#> [1] -0.00188636

```

## Doob $\chi^2$ family

The MRM framework’s “Doob  $\chi^2$  family” is a coordinated set of chi-square tests on the published Sprott / Doob / Iftene contingency tables (federal SIU operation, COVID-period operation, torture-classification rates, IEDM analyses). MORIE reproduces all five published statistics to within 0.01 of the printed values; details are in the MRM paper (`citation("morie")`).

## Where to go next

- The `effect-sizes` vignette covers Cohen’s d, Cramer’s V, omega-squared, proportion-CI, and the E-value in one place.
- The `mrn-otis-walkthrough` vignette shows the chi-square family applied to OTIS provincial data.